

How robust are linguistic markers of aging? The case of aging-related social media text

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ABSTRACT

Recent research suggests that linguistic biomarkers are effective in detecting probable cognitive impairment (PCI). This study extends this line of research by examining whether linguistic markers, which have been previously validated in structured, controlled clinical settings, are sufficiently robust to effectively detect PCI in unstructured, spontaneous, everyday social media posts. We hypothesized that classic linguistic markers, along with novel large language model (LLM)-derived measures, can robustly, reliably differentiate PCI from healthy controls (HC) based on social media posts. Both clinical interviews and social media posts were preprocessed to extract relevant content. We constructed natural language processing (NLP) pipelines to compute linguistic markers, and used LLMs to derive text similarity and surprisal metrics. Machine learning (ML) methods with explicit linguistic features were then used for predictive modeling. Additionally, we prompted LLM without explicit linguistic features to infer and detect PCI using zero-shot method. ML classifiers, especially gradient boosting, achieved high precision, recall, and overall accuracy in distinguishing PCI from HC across both data types (i.e., clinical interviews and social media posts). We also found that LLM-derived features and prompting methods provided moderate additional insights, and classic linguistic markers remained the most influential predictors. These findings demonstrated a reliable approach to transfer linguistic markers from clinical, controlled settings to spontaneous, unstructured social media posts. This study adds to the growing evidence that robust, AI-enhanced linguistic markers can advance early, noninvasive detection for cognitive decline in everyday language settings of social media posts.

1. Background and objectives

Recent advancements in natural language analysis have enabled the quantification of aging-related symptom severity through linguistic biomarkers, providing insights into how these symptoms appear in real-world settings and daily conversations (Bach et al., 2018; Chu et al., 2022; Raiaan et al., 2024). A linguistic biomarker is defined as a feature or set of features that correlates with a clinical outcome (Grabb and Brady, 2024). Such biomarkers can be used either to detect a pathological state or to monitor and classify its severity and stages of impairment (Corcoran et al., 2020; Gan et al., 2024; Javed et al., 2023; Tang et al., 2023). In computational terms, the search for linguistic biomarkers essentially involves identifying features that objectively

quantify language-related changes in various aging-related impairments and disorders.

Previous research has shown the effectiveness of linguistic biomarkers in detecting early signs of cognitive impairment using clinical data. However, as far as our knowledge goes, no studies have examined the degree to which these markers can be effectively transferred to the analysis of aging-related social media posts. This knowledge gap is significant as clinical interview data is costly to collect and can be bounded in content, while social media posts are widely available at minimal cost and they are rich in content. We explore and pinpoint the extent to which linguistic biomarkers can be used to robustly detect early signs of cognitive impairment from Facebook social media posts.

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1.1. Classic linguistic biomarkers of aging

There are multiple ways to extract and measure linguistic markers, each potentially offering informative and predictive insights of aging. Our study is inspired by and built up on previous work on classic linguistic markers of aging, where they assessed the cognitive status of older adults with probable cognitive impairment (PCI) through language. These linguistic markers, typically extracted from curated language data, have been used to evaluate two key domains of cognitive functioning: executive functioning and language/verbal skills, both of which have been effectively assessed through textual analysis methods (Anderson, 2005; Harvey, 2019).

Executive functioning was proxied by reasoning and measured using two indices, *sentiment analysis* and *semantic similarity*, which are widely recognized as NLP diagnostic markers for cognitive impairment (Nicholas et al., 1985; Tomoeda et al., 1996; Scibelli et al., 2018; Masrani et al., 2017; Morales et al., 2017; Hutto and Gilbert, 2014). *Verbal skills* were proxied by fluency, which was quantified through three indices: *part-of-speech* (POS), *lexical diversity*, and *syntactic complexity* (Fraser et al., 2015, 2019; Yancheva et al., 2015; Beltrami et al., 2018; Bucks et al., 2000; Ahmed et al., 2013; Jarrold et al., 2014; Ash et al., 2013; Budson and Kowall, 2013; Orimaye et al., 2014; Pakhomov et al., 2010).

Previous studies have established strong associations of cognitive status with these linguistic markers, primarily based on clinical data. Pertaining to our investigation, for *sentiment analysis*, studies showed that shifts in cognitive states can be detected by analyzing sentiment. For instance, studies have noted a higher frequency of negative words and a decreased use of positive words in individuals with (mild) cognitive impairment (MCI) (Scibelli, 2018; Morales et al., 2017; Lam et al., 2014; McIntyre et al., 2015; Yeung et al., 2021; Lindsay et al., 2021).

For *semantic similarity*, studies have validated that Alzheimer's disease (AD) and Alzheimer's disease and related dementias (ADRD) patients exhibit more empty speech, less contentful text, and increased text similarities due to the frequent repetition of specific words, phrases, and sentences (Nicholas et al., 1985; Tomoeda et al., 1996; Yeung et al., 2021; Lindsay et al., 2021).

For *part-of-speech*, previous studies indicated that changes in the proportion of different parts of speech have clinical significance. For example, an increased proportion of pronouns, adjectives, and verbs, coupled with a decreased proportion of nouns, has been found in ADRD (Fraser et al., 2015, 2019; Bucks et al., 2000; Ahmed et al., 2013; Jarrold et al., 2014). This overreliance on pronouns, which reflects an underlying difficulty in recalling specific nouns, and the concurrent overuse of adjectives and verbs are indicative of impairments in memory and lexical retrieval. Further, Huang et al. (2024) and Mueller et al. (2018) reported that content words such as verbs and adjectives were informative of PCI, and there is decrease in verb-to-adjective ratio in PCI compared to healthy older adults, reflecting difficulties with complex language production involving verbal predicates.

For *lexical diversity*, investigations suggested that vocabulary richness can serve as an indicator of cognitive health, with declines noted in ADRD and MCI (Fraser et al., 2015, 2019; Yancheva et al., 2015; Beltrami et al., 2018). This decline, which reflects a reliance on a smaller vocabulary set, aligns with other signs of cognitive deterioration (Lee, 2023).

For *syntactic complexity*, studies on complexity and sophistication suggested that analysis of syntactic and semantic structures can reveal difficulties related to memory and overall cognitive function. Clinical studies have validated a decrease of syntactic complexity in AD/ADRD (Fraser et al., 2015, 2019; Yancheva et al., 2015; Orimaye et al., 2014; Pakhomov et al., 2010) and aphasia (Ash et al., 2013). Reduced syntactic complexity, for example, fewer subordinate clauses or simpler sentence structures, has been observed in individuals with AD/ADRD and MCI, suggesting deficits in memory, attention, and language processing speed (Roark et al., 2007; Lee, 2023).

1.2. NLP analyses of aging-related social media text

NLP techniques commonly derived from non-clinical social media data, such as clustering and content analysis, have been used to examine the linguistic characteristics of individuals with dementia or mental health conditions. However, the application of these methods in the robust detection of cognitive impairment has been less frequent. Our major inspiration is from Thorstad and Wolff (2019), which showed that words derived from nonclinical subreddits could predict future postings to clinical subreddits, implying that everyday social media language contains signals of future mental illness. Their content analysis methods on social media clustering for mental health detection and at-risk situations guided our approach to extracting meaningful patterns from the Facebook data. Relatedly, Gkotsis et al. (2020) utilized a text analysis pipeline to analyze Reddit posts concerning dementia diagnoses, using automated topic gathering and clustering techniques to identify the attributes of the posts and their producers. Their NLP findings reveal common themes and concerns expressed in communications. Moreover, the work of Parsa et al. (2021) used various machine learning (ML) classifiers to assess the cognitive status of individuals based on their social media language.

Our study used social media data from Facebook for the following motivations. Previous studies have provided insights in leveraging Reddit, Twitter, as well as Facebook data to identify linguistic markers related to cognitive changes (Hrinu et al., 2022a, b; Thorstad and Wolff, 2019; Bacsu et al., 2022; Yoon, 2016; Oscar et al., 2017). Reddit tends to host discussions that are relatively topic-focused and less personal. In contrast, Facebook posts are typically more reflective of producers' real-life interactions and family dynamics. Eichstaedt et al. (2018) demonstrated that language use on Facebook is closely tied to personal emotional states and can even predict clinical conditions such as depression. The richer, family-oriented content available on Facebook offers a promising alternative for studies aiming to track aging-related symptom severity and real-world behavioral changes. We maintain that with Facebook posts, researchers can harness a more authentic snapshot of everyday language that is deeply embedded in producers' family and social contexts. Such data not only capture the nuanced expressions of individual experiences but also provide an ecologically valid basis for developing sensitive digital linguistic biomarkers.

As far as our knowledge goes, there are no NLP studies using aging-related social media text that particularly focused on the aforementioned linguistic markers. This gap suggests both research opportunity and challenge. Linguistic markers might manifest differently with different modality: from speech transcripts, written narratives, to social media text. Much of the prior work on linguistic biomarkers for cognitive impairment has relied on speech transcripts, which differ from social media posts in spontaneity, structure, and cognitive demands. Spoken language often reflects real-time processing, while written posts may be edited, curated, or assisted by spell-check tools. As such, the applicability of speech-derived markers to written language remains uncertain.

To address this challenge, we draw insights from previous work by Meier et al. (2024), which analyzed data from a diverse sample of participants who wrote personal narratives about their thoughts and feelings regarding healthy aging. Their findings suggested that language-based measures, in written narratives modality, may still offer benefits, capturing the less conscious aspects of well-being. Further, we curated a comparable dataset from clinical sites' speech transcripts, which includes the modality of the speech. This dataset demonstrates which linguistic markers remain robust in both spoken and written formats. As a proof of concept, we hope to provide an approach for assessing the robustness of linguistic markers across modalities. We maintain that this approach will showcase how to reliably utilize the rich, distinct, and naturalistic text data found in social media posts.

1.3. Aims and overall hypotheses

Our primary goal is to explore and pinpoint the extent to which linguistic biomarkers can be used to robustly infer PCI from social media posts. To address this overarching aim, we focused on two specific key questions.

First, how robust, stable are linguistic markers of aging, and specifically how do PCI and HC differ in linguistic markers in their social media posts? We sought to replicate and extend previous findings about linguistic markers sourced from clinical data. Previous studies, including Tang et al. (2023), have demonstrated that linguistic impairments are elevated among older adults and can even reveal subthreshold cognitive disturbances, suggesting that NLP features hold promise as accurate, noninvasive, and efficient biomarkers for such conditions. Building on this line of work, we hypothesize that linguistic markers identified in (clinically) controlled settings are transferable to spontaneous, unstructured social media textual data. In clinical sites, data collection is typically conducted in a hospital site or laboratory, where language is spoken then gets transcribed carefully, whereas social media posts are in free form. We maintain that demonstrating linguistic markers' stability, robustness and transferability would thus be an important advance. Besides the classic linguistic markers, we included recent AI systems to derive and validate new PCI detection techniques.

Our second key question focuses on the nature of these linguistic biomarkers: What specific linguistic markers are most predictive and sensitive of PCI? By developing predictive models, analyzing feature importance, and conducting simple individual statistical tests, we aim not only to enhance detection accuracy but also to better our understanding of how cognitive changes manifest in everyday social media language about aging. Given that cognitive decline can affect multiple aspects of language, following Thorstad and Wolff (2019), we hypothesize that even in everyday social media discussions on aging-related topics such as exercise, gathering, or nutrition and food, interpretable and stable linguistic markers indicative of impairment will emerge. Identifying such biomarkers across both clinical and social media datasets is crucial for developing precise, sensitive early screening tools for cognitive decline.

2. Methods

We proposed to address these two questions by leveraging NLP and ML techniques, where cognitive status measurement was bridged with specific linguistic markers, grounded in controlled, structured (with experimenters' pre-set linguistic tasks and discourse structures) and spontaneous, unstructured (without any pre-set linguistic tasks and discourse structures) data. We assessed *executive functioning and reasoning* via sentence similarity and sentiment analysis, revealing patterns of text similarity, repetition, and frequency of negative sentiments in PCI, while evaluating *verbal skills* through measures of syntactic complexity, part-of-speech distribution, and lexical diversity, which are

known to decline in PCI.

Concretely, first, we analyzed unstructured, spontaneous text data from social media posts, focusing on Facebook communities discussing Alzheimer's disease and related dementia symptoms and a corresponding healthy aging group discussing general health.

Second, clinically validated NLP markers were incorporated as benchmarks: We analyzed structured, controlled clinical datasets as an interpretation baseline, thereby enhancing our models' interpretability and reliability.

Third, we are aware of the rapid development of LLMs. We therefore computed LLMs-surprisals, in addition to the transformer LMs related measures of text similarity, and we prompted LLMs to automatically label a social media post as PCI or HC.

In Table 1, we summarized the linguistic markers we studied and their association with aging. In what follows, we presented the essentials of our datasets and experiments.

2.1. Data extraction and pre-processing

2.1.1. Group identification

We identified two primary groups PCI and HC for two data types (social media posts and clinical archived data). First, the publicly downloadable Meta Content Library data (Meta Platforms, Inc., 2024). "Downloadable public data" here refer to posts made by widely known individuals and organizations with a verified badge or 25,000 or more followers, and they can be downloaded as CSV files. Second, a clinical archive *DementiaBank* (Lanzi et al., 2023; Liu et al., 2023; MacWhinney et al., 2011; Macwhinney and Fromm, 2023; He et al., 2023; 2024), consisting of transcribed speech from a picture description task by mild and moderate PCI participants (combined due to small sample sizes) and healthy older adults controls. In particular, the task concerns a scene of *Cookie Theft*, where participants are instructed to describe what is going on in the Cookie Theft picture. We chose this task, given the consideration that it is available for a large number of participants, the task is controlled and rarely varies across clinical sites, and it minimizes episodic memory demands and produces structured clinical narratives (He et al., 2023; 2024; Lofgren and Hinzen, 2022).

Although our focus was on aging-related social media posts, we included a validation dataset collected from clinical sites. This is to systematically demonstrate how the proposed linguistic markers behave in carefully structured, clinically controlled data, and to examine the markers' stability across different data formats to identify transferable markers from clinical settings to social media. We iteratively refined our NLP models by testing them on these labeled clinical datasets that included cognitive health status. Demographics summary for all the datasets was given in Table 2.

We acknowledge that PCI and HC data in social media posts might not be directly comparable to those in clinical sites, which were transcribed clinician-participant's interviews. To mitigate these differences, we implemented several strategies, including controlling text length,

Table 1

An overview of linguistic markers studied in our investigation about aging-related social media text.

Linguistic marker	Characterization	Cognitive domain assessed	Association with cognitive impairment
Sentiment	Use of positive/negative affect words	Executive functioning (reasoning)	Increased negativity and reduced positivity in language are linked to emotional dysregulation and cognitive decline in PCI and MCI populations.
Semantic similarity	Redundancy and lexical repetition	Executive functioning (reasoning)	Higher similarity reflects reduced content richness and idea generation, common in Alzheimer's and related dementias.
Part-of-Speech (POS)	Proportion of nouns, pronouns, adjectives, verbs	Verbal skills (lexical retrieval)	Overuse of pronouns and underuse of nouns suggests difficulties with specific word retrieval; reduced verb-to-adjective ratio reflects struggles in producing complex language.
Lexical diversity (TTR)	Variety of vocabulary used	Verbal skills (Fluency)	Lower lexical diversity reflects reliance on a limited vocabulary, indicating cognitive decline in memory and fluency.
Syntactic complexity	Sentence length and subordination	Verbal skills (Complexity, Memory)	Simpler sentence structures (e.g., fewer subordinate clauses) indicate reduced cognitive resources for processing and planning complex syntax.
LLM-based surprisal	Predictive likelihood of next word (surprisingness)	Both executive function and verbal skills	Higher surprisal values suggest less predictable, less grammatically coherent text, reflecting reduced planning, memory, and coherence in individuals with cognitive impairment.

Table 2

Producers and participants demographics summary. All data were pre-processed and matched in text length within and between groups. Texts with outlier length were excluded.

Meta	Total number of posts	Posts identification (search keywords)
PCI	473	Alzheimer's Society, Dementia, ADRD, Alzheimer's Awareness, Alzheimer's and Dementia Support Group.
HC	473	AARP (<i>American Association of Retired Persons</i>), a Facebook interest group discussing issues influencing those 50 and older). Aging Healthy, American Society on Aging, Aging. Clinical site
DementiaBank	Total number of utterances	
PCI	104 (104, mean age 71)	Protocol Baycrest and Protocol Delaware with MCI, Non-protocol Pitt with probable AD.
HC	104 (104, mean age 65)	Protocol Baycrest and Protocol Delaware, Non-protocol Pitt.

selecting consistent linguistic features, and systematically preprocessing texts to remove disfluencies and irrelevant content.

We also acknowledge that there is uncertainty regarding post authorship and content in our social media dataset. In support groups such as “Alzheimer’s and Dementia Support Group” in Table 2, caregivers or family members may post on behalf of individuals with cognitive impairment, particularly in advanced stages. Moreover, like many other social media platforms, Facebook posts contain text for promotion and advertisement. To address these uncertainties, first, we highlighted that previous work (c.f., Section 1.1 and 1.2: Yoon, 2016; Hrinu et al., 2022a, b; Thorstad and Wolff, 2019; Bacsu et al., 2022; Oscar et al., 2017) identified posts mentioning dementia-related keywords and inferred PCI accordingly. This method proved effective in helping researchers gain insights for dementia care in social media texts. This line of work justified our approach in identifying and determining the PCI group.

Second, we further removed noise such as advertisements and promotional content using an iterative pipeline involving clustering, ensuring analysis of only relevant social media posts. We preprocessed 2680 PCI and 3071 HC posts, and they were reduced to 473 each. Clinical data from DementiaBank underwent CHAT-format cleaning while preserving verbatim content, but clustering was not performed due to its structured discourse. The pre-processed text from the picture description task was aggregated at the speaker level to match the approach used for social media posts. Both PCI and HC utterances were reduced to 104 each.

2.1.2. Data pre-processing details

We first preprocessed our datasets, which contained 2680 posts produced by PCI and 3071 posts produced by HC. We cleaned text by removing Uniform Resource Locators (URLs), hashtags, emojis, and converting all text to lowercase, which was consistent and computable for NLP tools. Second, we transformed all textual data into numerical vectors using TF-IDF (Term Frequency-Inverse Document Frequency), which highlights words that are relatively more important in each document compared to their overall frequency in the corpus.

To identify how many clusters would best represent these TF-IDF vectors, we used two established heuristics: the Elbow method and the Silhouette score (Martin and Jurafsky, 2009). The Elbow method involves plotting the sum of squared distances (SSE) between each data point and its nearest cluster centroid as the number of clusters increases; the point where the SSE curve begins to level off (“the elbow”) often suggests an optimal number of clusters. In contrast, the Silhouette score measures how similar each data point is to its own cluster compared to other clusters, producing a value between -1 and 1 ; higher scores indicate better-defined clusters. This analysis suggested that 9 clusters

provided the highest separation. Given the Silhouette score’s focus on cluster quality, we proceeded with 9 clusters using K-means, a centroid-based algorithm that partitions data, so each point is assigned to the cluster with the nearest centroid. To visualize the already identified clusters, we then used t-SNE (t-distributed Stochastic Neighbor Embedding), a dimensionality reduction technique. We visualized these high-dimensional TF-IDF vectors in two dimensions, where each point represents a post, and clusters appear as distinct color groupings. By inspecting the top 10 keywords extracted from TF-IDF-weighted cluster centroids and representative post instances, we interpreted the thematic content of each cluster. Clusters dominated by third-person accounts, informational caregiving content, or institutional research updates were excluded, refining the dataset to ensure alignment with our aging-related research objectives. Cluster themes and exclusion criteria are summarized in Supplementary Table S2.

To control for text length differences between HC and PCI groups, we applied a corpus-level length matching procedure. We first computed the total word count for each group. The group with the smaller count (PCI) was the reference. Posts from the larger group (HC) were sequentially accumulated until their cumulative word count matched the reference. If the final post exceeded the target, it was truncated at the point where the total word count matched the reference. The resulting subset was mapped back to the original post structure for analyses. This approach matched the groups in overall text length while minimizing data distortion. Unlike IQR-based or post-level length filtering, which may disproportionately remove long narrative posts that are especially common in social media, corpus-level length matching preserves the natural post length distribution and affects only a single post at the truncation boundary. This method is well suited for comparing heterogeneous text sources, including clinical narratives and social media posts, where long-form descriptions may be analytically meaningful. Therefore, no fixed post-level length threshold was applied; instead, the effective threshold was defined by the total word count of the smaller corpus. We applied the same quality- and length-control procedures for HC posts. After preprocessing and text matching, the PCI group posts reduced from 2680 to 473, and the HC group posts reduced from 3071 to 473.

The preprocessing of clinical data from the DementiaBank started with orthographically transcribed text, with detailed annotation. In the CHAT format used by CLAN (Computerized Language Analysis) software, “& = laughs” indicates non-speech verbalizations like laughter, while “[+ exc]” marks extraneous comments such as “wait a second”. These codes instruct the software to exclude the associated utterances from analysis. As a result, a CHAT-annotated transcription appears as: “& = laughs I haven’t really had an injury. So [+ exc] luckily I haven’t had any injuries”. We cleaned the CHAT-formatted text by removing non-alphabetic symbols while preserving its verbatim content, in line with how social media posts were maintained. This means that we kept transcripts’ replacements of the participants’ words, repeated words were removed so only one instance remains, partial words spoken by the participant were made full, and nonwords were removed. Filler words were still kept. As a consequence, the pre-processed CHAT formatted text became: “I haven’t really had an injury. So luckily I haven’t had any injuries”. We did not perform clustering analysis for the DementiaBank data, because its discourse was structured and its content was bounded. Finally, pre-processed text from the picture description task was aggregated at the speaker level to highlight individual linguistic differences, matching the approach used for social media posts.

2.2. Linguistic markers extraction and computation

We computed a range of linguistic markers to detect differences between the PCI and HC groups, covering sentiment, semantic similarity, part-of-speech (POS) patterns, lexical diversity, syntactic complexity, and LLM-derived surprisal.

Sentiment analysis was performed using two complementary

approaches. First, we applied the VADER sentiment analyzer (Hutto and Gilbert, 2014) via the NLTK toolkit (Bird, 2006) to compute sentence-level positive, negative, and neutral scores, aggregating a compound score for each post. Second, we used a dictionary-based method (Bradley and Lang, 1999; Warriner et al., 2013) to extract valence, arousal, and dominance ratings on a scale from 1 (lowest) to 9 (highest). These dictionary-based values were converted to z-scores and normalized by text length. The combination of methods allowed detection of emotional tone shifts, with the hypothesis that PCI posts would exhibit lower sentiment scores than HC, reflecting potential executive functioning decline.

Semantic similarity was measured using SentenceTransformer embeddings from the all-mpnet-base-v2 model (HuggingFace ID: sentence-transformers/all-mpnet-base-v2; Masrani et al., 2017). Embeddings were generated either by averaging word vectors or applying TF-IDF weighting, and cosine similarity between adjacent utterances was computed to assess textual repetition and redundancy. Higher similarity and repetition were hypothesized for PCI, consistent with perseveration.

POS indices were calculated by tagging all words and normalizing each category's frequency by the total word count. We computed NOUN%, PRON%, NOUN_VERB, PRON_NOUN, Verb_Adj, and Noun_to_Adj ratios to capture grammatical balance and shifts in lexical choice. We predicted that PCI would show lower NOUN%, NOUN_VERB, and Verb_Adj ratios, but higher PRON% and PRON_NOUN ratios, reflecting lexical simplification.

Lexical diversity was quantified with two measures: type-token ratio (TTR), calculated as the number of unique word types divided by total tokens, and semantic diversity (SemD), derived from the same dictionaries used in sentiment analysis. TTR is a text-level measure of lexical variety, while SemD is a word-level measure of contextual variability, capturing generality versus specificity. We hypothesized that PCI would have lower TTR but higher SemD, suggesting reliance on general-purpose vocabulary.

Syntactic complexity markers included structural and functional indices derived using spaCy (https://spacy.io/) and NLTK (Wagner et al., 2010). We computed total word and sentence counts, normalized dependents, coordinate phrases, and complex nominals. Subordination was measured as dependents per clause, coordination as coordinate phrases per clause, and phrasal sophistication as complex nominals per clause, all normalized by text length. PCI was expected to show reduced syntactic complexity.

Furthermore, we explored surprisal features derived from pre-trained autoregressive transformer large language models (LLMs, Wolf et al., 2020). These models utilize causal language modeling, a pre-training approach where the model processes text sequentially, predicting the next word based only on the preceding context. This makes them well suited for calculating surprisal, which conceptually captures text "surprisingness" and predictiveness given preceding contexts. Widely used in information theory as a classic index and recently gaining renewed attention in LLMs research, surprisal is mathematically defined as the negative log-probability of a word's occurrence in an utterance given its prior context (Levy, 2008; Radford et al., 2018). Previous clinical studies have shown that LLMs-surprisal tends to increase in (older) adults with aphasia, likely due to the decrease of grammaticality, hence less predictive and more "surprising" language production in older adults (Rezaii et al., 2021, 2023, 2024; Rezaii, Michaelov, et al., 2023; Cong et al., 2024). In the context of categorizing early signs of PCI, we propose that LLM-surprisals capture multi-facets of cognitive status, possibly involving both executive functioning and verbal skills.

Specifically, LLM-derived surprisal was calculated as the negative log-probability of each token given its preceding context, summing probabilities for multi-token words. Utterance-level surprisal was normalized by length, and paragraph-level surprisal was computed using mean, median, standard deviation, minimum, and maximum values across utterances within each participant's response. To assess model scaling effects, we used two decoder-only, causal language

models: GPT-2 (124M parameters; Radford et al., 2019) and Mistral-7B (v0.1, 7B parameters; Mistral AI Team, https://mistral.ai/). Surprisal calculations were implemented using the open source minicons library (Misra, 2022). Higher surprisal values were hypothesized for PCI, indicating diminished grammaticality and sentence predictability. All models used in these computations were open-source and accessed through HuggingFace (https://huggingface.co/models) as of February 2025.

2.3. Model construction and optimization

To classify PCI versus HC, we trained and optimized four machine learning models: decision tree, random forest, gradient boosting, and support vector machine (SVM), each offering distinct computational advantages. Decision trees partition the feature space, random forests aggregate predictions from multiple trees, gradient boosting iteratively corrects errors from prior iterations, and SVM finds an optimal separating hyperplane between classes. This diverse modeling strategy enabled testing of the predictive power of linguistic and LLM-based markers across multiple algorithmic frameworks.

Feature selection was performed by removing variables with Pearson correlation coefficients above 0.65 to reduce redundancy. The retained features included the following. Sentiment metrics: VADER compound score, valence, and arousal; Semantic similarity: cosine similarity; POS features: PRON%, NOUN%, NOUN_VERB, PRON_NOUN, Verb_Adj, and Noun_to_Adj; Lexical diversity: type-token ratio (TTR) and semantic diversity (SemD); Syntactic complexity: dependents per clause, coordinate phrases per clause, complex nominals per clause, total dependents, and total complex nominals; LLM-derived surprisal: mean surprisal from GPT-2 and median surprisal from Mistral-7B, with the median used for Mistral due to skewness in its surprisal distribution.

Training and evaluation followed a validated framework to prevent overfitting. The dataset was split into training (67%) and hold-out test (33%) sets, equivalent to a two-thirds/one-third division. We applied nested 3-fold cross-validation with grid search to tune hyperparameters and mitigate overfitting, following best practices recommended by Cawley and Talbot (2010). Hyperparameter tuning was performed only within the inner folds, while final evaluation was conducted exclusively on the unseen test set to ensure an unbiased assessment of model performance.

Model-specific configurations were as follows: For the clinical dataset, the best-performing model was a Gradient Boosting Classifier with $n_{\text{estimators}} = 300$, $\text{criterion} = \text{friedman_mse}$, and $\text{max_features} = 19$. For the social media dataset, the best-performing model was also a Gradient Boosting Classifier, but with $n_{\text{estimators}} = 100$, $\text{criterion} = \text{friedman_mse}$, and $\text{max_features} = 5$. In both datasets, class 0 represented HC and class 1 represented PCI.

Model interpretability was addressed using SHAP (SHapley Additive exPlanations) values, applied to the best model configurations to quantify individual feature contributions and provide insight into prediction drivers.

2.4. With zero linguistic markers, how would LLMs perform in detecting PCI?

To evaluate LLMs' performance without any structured linguistic inputs, we implemented zero-shot prompting. In this setup, the model receives a prompt and a raw social media post, without being trained or fine-tuned on labeled data. We sampled 200 posts each from the PCI and HC datasets used in earlier experiments. To ensure length comparability, we truncated PCI posts exceeding 36 words, the mean length of HC posts, yielding a final dataset of 200 PCI posts with a mean length of 35.95, and 200 HC posts with a mean length of 27.31.

We used Llama-3-8b (8 billion parameters; LLaMA 3 Model Card, AI@Meta, 2024) with the prompt: "Return 1 if the post '{topic}' is produced by a person who is in the probable cognitive impairment group and

O otherwise". Here, "{topic}" was replaced with the full text of the post. Following the procedure outlined by Qiu et al. (2024), we used a one-trial-per-run method in which each session involved only a single post and corresponding prompt. This approach prevented carryover effects across prompts and minimized context contamination, while also reducing susceptibility to connectivity or memory loss issues. We used the default settings for Llama-3-8b and calculated precision, recall, and accuracy on the binary output labels returned by the model.

Such a minimal prompt was used, because this served our goal to evaluate the extent to which a general purpose, pretrained LLM, without task specific guidance, can implicitly capture PCI related linguistic signals that have been reported in prior literature. By avoiding explicit decision criteria, we aimed to assess the LLM's zero-shot sensitivity rather than its ability to follow explicit linguistic features or rules. We intentionally designed this setting to reflect a realistic deployment scenario, in which labeled data or domain expert defined criteria may be limited. We acknowledge that a more structured prompt could potentially improve performance and interpretability. Therefore, as an additional exploration, we tested the following structured prompt variant, and designed in such a way that LLM used it as the basis for classification:

You are given a social media post and must classify it as PCI (probable cognitive decline) or HC (healthy control) using linguistic evidence. Evaluate whether the post exhibits aging-related linguistic markers associated with cognitive impairment: increased negative and reduced positive sentiment; high semantic similarity due to redundancy or lexical repetition indicating reduced content richness; part-of-speech patterns such as overuse of pronouns, underuse of nouns, or a reduced verb-to-adjective ratio suggesting word-retrieval and syntax difficulties; low lexical diversity reflecting reliance on a limited vocabulary; reduced syntactic complexity characterized by shorter sentences and fewer subordinate clauses; and elevated LLM-based surprisal indicating less predictable or less grammatically coherent language. If the post shows multiple such patterns, classify it as PCI and return 1; otherwise classify it as HC and return 0. Output only 1 or 0, with no explanation.

The same model settings and prompting pipelines used in the original prompting task was applied to this additional experiment with this prompt variant.

3. Results

3.1. Linguistic markers for aging in clinical sites

Table 3 showed that overall, for clinical datasets, the gradient boosting classifier gave better predictions than other classifiers, with an accuracy, precision, recall, F1-score of 0.91. Particularly for detecting the positive class of PCI, the gradient boosting classifier achieved 0.91 in weighted average precision, recall, and F1-score. SVM showed an accuracy of 0.88, a low performance in detecting all positive instances in

the dataset (recall = 0.88), and a high performance in precisely detecting the positive class PCI (precision = 0.91). When detecting the PCI class, interestingly, SVM gave a precision of 1, suggesting that every text the model predicted as PCI was indeed a PCI case, and there were no false positives. On the other hand, a recall of 0.76 indicated that SVM correctly identified 76% of all actual PCI text, meaning that 24% of true PCI cases were missed. We infer that this trade-off and very conservative positive predictions can occur when the clinical dataset is small, as SVM often requires larger sample sizes. Prior work on text categorization has similarly noted that SVM can be overly selective with small training samples, leading to high precision but lower recall (Santos, 2024; Matykiewicz and Pestian, 2012).

Decision tree and random forest classifiers did not show better performance in predicting PCI and HC in clinical data, with an accuracy of 0.87 and 0.91 respectively. When detecting the PCI class, similar to the SVM, the random forest classifier was also able to precisely detect PCI (precision = 0.94) but not as good at recalling all PCI instances in the dataset (recall = 0.88). Decision tree showed the inverse pattern, with a higher recall and a lower precision. Our interpretation was that the decision tree may have over-fit to noise in the training set, which can lead to higher recall, catching most true positives, but lower precision and more false positives, because they may use overly complex or unstable decision rules. This inverse pattern compared to SVM seems typical for unpruned decision trees in small sample settings. We did not conduct any pruning of the trees given our small datasets, which could have potentially improved the performance of more interpretable decision trees.

Overall, the four classifiers showed decent performance, suggesting that the linguistic markers were informative in detecting language users' cognitive status, effectively teasing apart PCI and HC. In the clinical field, ML metrics such as F1 scores exhibit values ranging from 0.66 to 0.96 (Hicks et al., 2022), where a value close to 1 represents good precision and recall values. These findings indicated that our interpretation baseline associating linguistic biomarkers with cognitive status (PCI, HC) was established.

We next visualized how each linguistic biomarker contributes to gradient boosting prediction using the SHAP value (Fig. 1), since the gradient boosting classifier (as well as the random forest) showed the best efficacy, and gradient boosting classifier indicated a more class balanced performance (Table 3). Our findings showed that lexical diversity indices (SemD and TTR) were the most decisive in separating PCI and HC, followed by POS indices (PRON%), and sentiment indices such as arousal and valence. POS indices, such as Pron_to_Noun, Noun_to_Adj, and Noun_to_Verb, and syntactic complexity indices total_dependents ranked relatively high in overall feature importance. Semantic similarity index average_similarity was in the middle of the feature importance rank, suggesting its moderate influence in predictive modeling. Interestingly, the two novel linguistic markers derived from LLMs, mistral_median_surp and gpt2_mean_surp, also showed moderate importance, despite their hypothetically sophisticated representation of language naturalness. In other words, LLMs-surprisals appear lower on the y-axis, indicating that they are less impactful compared to other direct features such as TTR, total_dependents, or PRON%. This finding suggested that new markers did not surpass classic ones.

In addition to the statistical significance of each linguistic marker's impact on the model's prediction, we also examined their direction. Specifically, negative SHAP values indicated a stronger association with class 0 HC, while positive SHAP values indicated a stronger association with class 1 PCI. For the two lexical diversity markers, SemD showed increase in PCI and a mixture of increase and decrease in HC; whereas TTR showed decrease in PCI and mostly increase in HC, as predicted. For the part-of-speech index PRON%, there was an increase in PCI and a mixture of increase and decrease in HC. For the two informative sentiment indices, both Arousal and Valence showed decrease in PCI and increase in HC, which aligned with our predictions.

Table 3

Predictive efficacy of the best configured model to predict PCI and HC in clinical sites.

	Precision	Recall	F1-score	Accuracy
Gradient boosting classifier weighted average	0.91	0.91	0.91	0.91
Gradient boosting detecting the PCI class	0.91	0.91	0.91	
SVM weighted average	0.91	0.88	0.88	0.88
SVM detecting the PCI class	1.0	0.76	0.86	
Decision tree classifier weighted average	0.87	0.87	0.87	0.87
Decision tree detecting the PCI class	0.83	0.91	0.87	
Random forest classifier weighted average	0.91	0.91	0.91	0.91
Random forest detecting the PCI class	0.94	0.88	0.91	

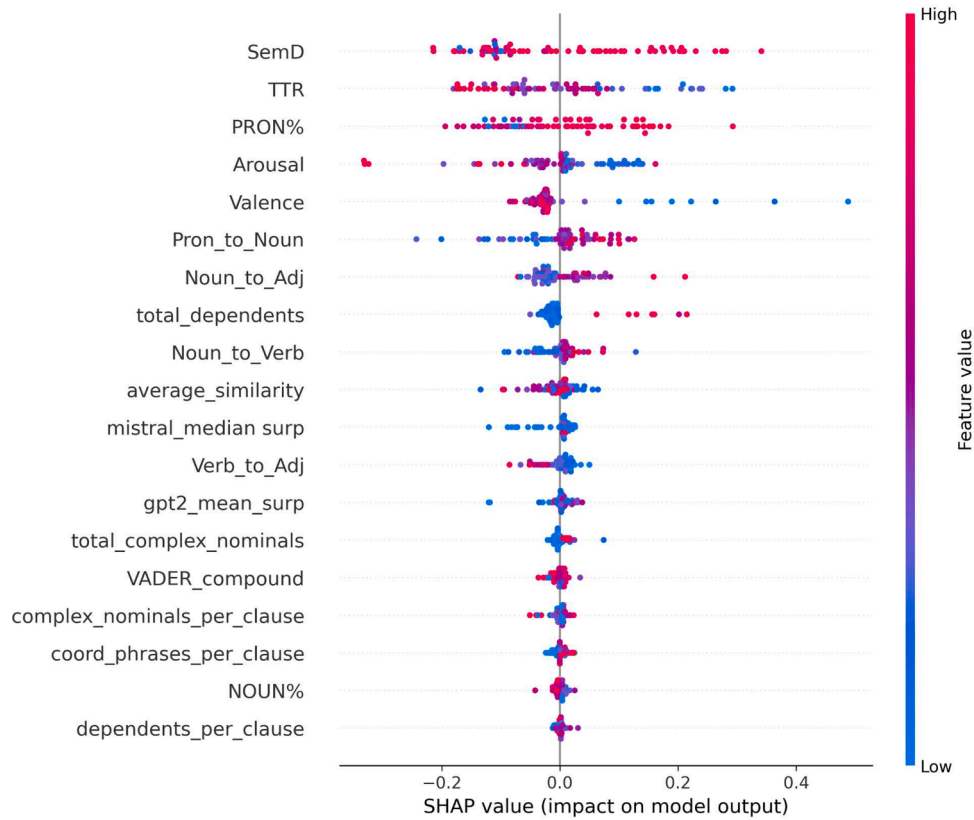


Fig. 1. Features importance rank for the best configured model predicting PCI and HC data sourced from the DementiaBank.

3.2. Linguistic markers for aging in social media posts

Table 4 summarizes classifier performance in distinguishing PCI from HC using social media posts. In general, models trained on social media data showed higher predictive performance than those trained on structured clinical data. This difference may be partially associated with the larger sample size. Notably, the two Facebook communities represent distinct populations, and differences in communication patterns might have influenced model performance. Such population-level differences could either introduce additional discriminative signals or confounding factors. These differences should be considered when interpreting the observed performance advantages of social media-based models.

In particular, the gradient boosting classifier achieved 0.96 in precision, recall, F1-score, and accuracy. Same scores were found in the random forest classifier. But the two classifiers differed in detecting PCI,

Table 4

Predictive efficacy of the best configured model in classifying PCI and HC in social media posts.

	Precision	Recall	F1-score	Accuracy
Gradient boosting classifier weighted average	0.96	0.96	0.96	0.96
Gradient boosting detecting the PCI class	0.97	0.95	0.96	
SVM weighted average	0.95	0.95	0.95	0.95
SVM detecting the PCI class	0.95	0.94	0.95	
Decision tree classifier weighted average	0.92	0.92	0.92	0.92
Decision tree detecting the PCI class	0.93	0.92	0.92	
Random forest classifier weighted average	0.96	0.96	0.96	0.96
Random forest detecting the PCI class	0.98	0.94	0.96	

with the gradient boosting classifier being slightly more balanced in precision and recall. The SVM classifier performed slightly lower overall, with 0.95 in precision, recall, F1-score, and accuracy. The decision tree classifier showed the lowest performance, achieving 0.92 in all metrics.

Overall, our findings showed that the linguistic markers validated in clinical datasets can be transferred and re-used reliably for predicting PCI in social media posts. Further, leveraging larger and more diverse datasets enhanced models' learning and predicting.

We next visualized how each linguistic biomarker contributes to gradient boosting prediction using the SHAP value (Fig. 2), since the gradient boosting classifier showed the best overall efficacy. Similar to the linguistic markers' pattern in clinical settings, lexical diversity index TTR was also the most impactful in social media posts. Syntax complexity and POS, such as total_dependents and NOUN%, were also informative in predicting PCI and HC, followed by sentiment indices Valence and VADER_compound. Other indices were similarly important, although not ranked as high, including syntactic complexity indices total_complex_nominals, complex_nominals_per_clause, coord_phrases_per_clause, and POS indices Pron_to_Noun, Noun_to_Verb, and PRON%. Semantic similarity index average_similarity showed moderate impact in predictive modeling. Moreover, the two LLM-derived naturalness indices, mistral_median_surp and gpt2_mean_surp, showed limited impact, as indicated by their low ranks compared to most linguistic markers (Fig. 2).

The directions of the linguistic markers' relationships between PCI and HC were mostly as predicted, with decrease in PCI and increase in HC for TTR, suggesting repetitive language and lexical decline in PCI relative to HC. Total_dependents showed decrease in HC and a mixture of decrease and increase in PCI. NOUN% showed an inverse pattern: decrease in PCI and increase in HC, as predicted. Valence showed mostly decrease in PCI and increase in HC, whereas VADER_compound showed the inverse, with some mixture of increase and decrease in HC.

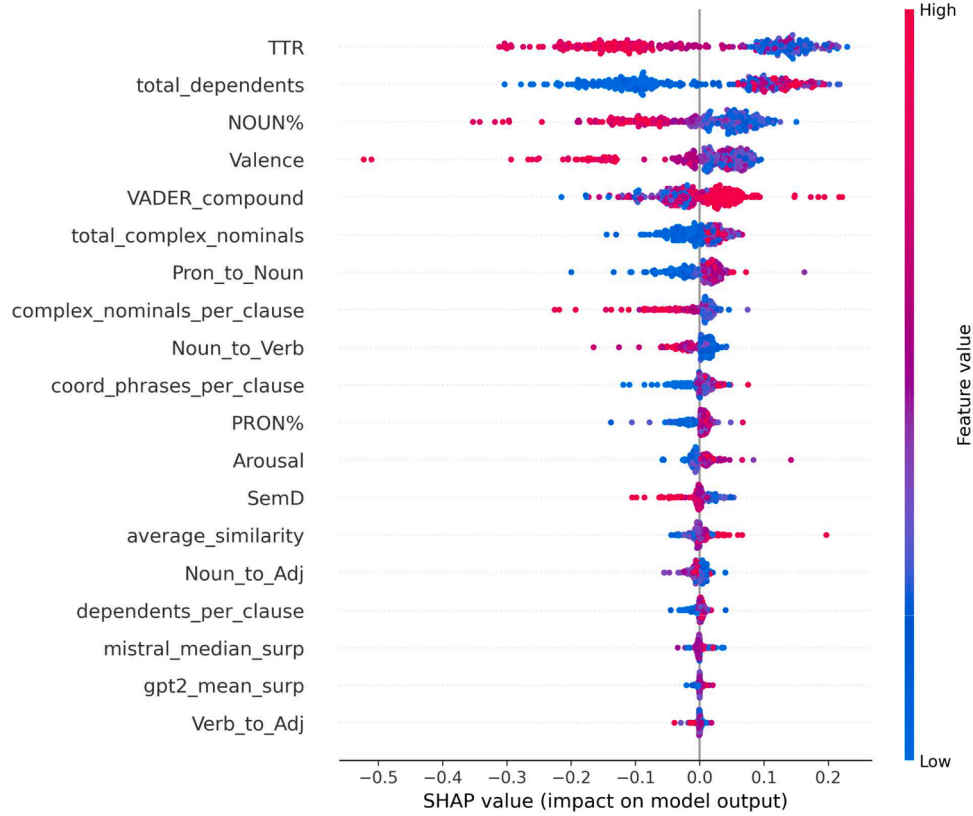


Fig. 2. Features importance rank for the best configured model predicting PCI and HC data sourced from the social media posts.

Why do important features diverge between clinical transcripts and social media posts? We gave some possible explanations. First, in the clinical setting, the importance of PRON% and SemD likely reflects characteristics of the picture description task. In this constrained environment, cognitive decline manifests as empty speech, namely a reliance on pronouns, possibly because the patient cannot retrieve specific nouns. Conversely, social media is relatively unconstrained and conversational. Here, syntactic complexity (e.g., total_dependents, complex_nominals) might become more discriminative. We speculate that cognitive impairment in social media is revealed by a simplification of sentence structure and a reduction in the density of information, rather than just noun retrieval failures. Second, the shift from arousal and valence in clinical sites to VADER compound in social media suggests that emotional intensity is a stronger signal in formal interviews (where patients may feel stressed), while tonal polarity, such as positive vs. negative, is more informative in the social, opinion sharing environment of Facebook.

3.3. A closer look at individual linguistic markers

Additionally, we explored each linguistic marker individually. We constructed Welch two sample T-test models for each individual linguistic marker, comparing their distribution in PCI and HC (Table 5). Our findings showed that sentiment markers aligned more with predictive modeling of linguistic markers in the clinical dataset than in the social media posts, POS markers mostly aligned with predictive modeling on social media posts, lexical diversity markers (TTR) generally aligned with predictive modeling on both clinical and social media posts, and that syntactic complexity markers (complex nominals per clause) echoed with predictive modeling on social media posts data, showing decrease in PCI.

We expected an increase in LLMs-surprisal scores for PCI. Interestingly, a statistically significant decrease was identified instead. This is

Table 5

Welch two sample T-test (assuming that two groups are not of equal variances) group comparison for each variable. Notation: ↓ indicates that PCI is lower than HC, namely a decrease in PCI relative to HC; ↑ = PCI increases, higher relative to HC, **bold** cells indicate variables for which both direction and statistical significance matched our predictions.

General category	Specific variables	DementiaBank	Facebook posts
Sentiment analysis	VADER-compound	↓ <i>p</i> = 0.041	↑ <i>p</i> < 0.001
	Valence	↓ <i>p</i> = 0.760	↓ <i>p</i> < 0.001
	Arousal	↓ <i>p</i> < 0.001	↓ <i>p</i> = 0.413
Semantic similarity	Cosine similarity	↓ <i>p</i> = 0.003	↑ <i>p</i> < 0.001
Part of speech (POS)	PRON%	↑ <i>p</i> < 0.001	↑ <i>p</i> < 0.001
	NOUN%	↓ <i>p</i> = 0.282	↓ <i>p</i> < 0.001
	NOUN_VERB	↓ <i>p</i> = 0.662	↓ <i>p</i> < 0.001
	PRON_NOUN	↑ <i>p</i> < 0.001	↑ <i>p</i> < 0.001
	Noun_to_Adj	↓ <i>p</i> = 0.069	↓ <i>p</i> = 0.023
Lexical diversity	Verb_to_Adj	↓ <i>p</i> = 0.003	↑ <i>p</i> < 0.001
	TTR	↓ <i>p</i> < 0.001	↓ <i>p</i> < 0.001
	SemD	↑ <i>p</i> < 0.001	↑ <i>p</i> < 0.001
Syntactic complexity	Dependents per clause	↑ <i>p</i> = 0.621	↑ <i>p</i> < 0.001
	Coordinate phrases per clause	↑ <i>p</i> = 0.118	↑ <i>p</i> < 0.001
	Complex nominals per clause	↓ <i>p</i> = 0.039	↓ <i>p</i> < 0.001
Text predictiveness	Total_dependents	↑ <i>p</i> < 0.001	↑ <i>p</i> < 0.001
	Total_complex_nominals	↑ <i>p</i> < 0.001	↑ <i>p</i> < 0.001
	Mean GPT2 surprisal scores	↑ <i>p</i> = 0.230	↓ <i>p</i> < 0.001
	Median Mistral surprisal scores	↓ <i>p</i> < 0.001	↓ <i>p</i> < 0.001

likely because linguistically and cognitively, posts focused on reassuring

or encouraging (e.g., posts involving emotional support content) may prioritize clarity and conventional phrasing over novelty, which can lower surprisal, independently of cognitive status. Moreover, PCI language might exhibit more restricted vocabulary, which can increase predictability for LLMs and lower the surprisal. Additionally, qualitatively, some PCI posts appeared to adopt highly standardized, informational, or advocacy-oriented language, which can reduce linguistic unpredictability for LLMs. For example, posts involving national diagnosis statistics use familiar public health discourse structures with predictable lexical choices, resulting in very low surprisal.

Also, we found that LLM-surprisals individually were able to distinguish PCI and HC. However, predictive modeling showed that LLM-surprisals were only moderately impactful when in combination with other linguistic markers (c.f. Fig. 2). It is possible that there may be overlaps or contrasts between what LLM-surprisals and what the classic indices were capturing. For example, we infer that since language in PCI was restricted in vocabulary, their surprisals were perhaps lower. We therefore conducted correlational analyses on LLM-surprisals, in order to interpret their relationships with lexical diversity indices. Pearson correlation showed that *mistral_median_surp* and TTR were positively correlated ($r = 0.203$; $p < 0.001$) in clinical datasets, and that the two were also positively correlated ($r = 0.182$; $p < 0.001$) in social media datasets. These correlations partially support our speculation of attributing lower LLM-surprisals in PCI to lower lexical diversity. We did not find significant correlations between *mistral_median_surp* and SemD in clinical datasets ($r = 0.01$; $p = 0.823$) or in social media posts ($r = -0.01$, $p = 0.823$).

We additionally conducted stability tests, where we reported Cohen's d in Table 6, as a quantitative measure of the standardized mean difference for each linguistic feature across two comparisons: HC_social vs. HC_clinical, and PCI_social vs. PCI_clinical. These values indicate the magnitude of the difference between the group means, in terms of their pooled standard deviation. Larger absolute Cohen's d values suggest a more substantial difference between the groups for that particular feature, while values closer to zero imply greater similarity and stability between the social and clinical contexts for that specific diagnostic group. More details about specific distribution descriptive statistics were provided in Supplementary Table S1.

Table 6 showed varying degrees of stability for linguistic features between social (Facebook posts) and clinical (DementiaBank transcripts) modalities. For healthy controls (HC DementiaBank vs. HC

Facebook posts), features like VADER_compound, valence, and total_complex_nominals show relatively small effect sizes (absolute $d < 0.2$), suggesting these markers are quite stable across different modalities for this group. On the other hand, when comparing PCI DementiaBank vs. PCI Facebook posts, many features, including valence, arousal, average_similarity, various POS proportions (PRON%, NOUN%), and syntactic complexity measures like dependents_per_clause and complex_nominals_per_clause, exhibit large effect sizes (absolute $d \geq 0.8$). This indicates significant differences in these linguistic markers depending on the modality for PCI. Similar large effect sizes for features like NOUN%, Pron_to_Noun, TTR, and SemD are also observed in the HC DementiaBank vs. HC Facebook posts comparison, highlighting that these features are generally not stable across modalities even in the HC group. Notably, *mistral_median_surp* shows a very large effect size in both comparisons, suggesting it is highly sensitive to the change in modality.

3.4. Zero-shot LLMs prompting

Our zero-shot prompting showed that Llama-3-8b achieved an accuracy of 0.784. It was able to precisely label most of the PCI cases (precision = 0.764), and it showed good performance in catching all the PCI cases (recall = 0.827). Quantitatively, when detecting PCI, Llama-3-8b was not as robust as most of the predictive models involving explicit structured linguistic markers: even the worst ML model, the decision tree classifier, already showed 0.87 in accuracy, precision, and recall.

Qualitatively, we illustrated a few responses from Llama-3-8b. We found that Llama-3-8b's reasoning for labeling posts of PCI appears inconsistent when relying solely on zero-shot prompting. For example, one response states, "Based on the language and content of the post, I would classify it as 1... the writer's thoughts are disjointed and unclear", suggesting a cognitive impairment based on perceived fragmentation and disorganization. In contrast, another response indicates, "I would classify it as 1 (probable cognitive impairment group). The post appears to be written in a clear and concise manner, with no grammatical errors or complex sentence structures. The vocabulary is also straightforward ...", leading to the same classification result despite evaluating different content. These contradictory assessments indicate the limitations of zero-shot prompting, where the model's reliance on surface-level cues can produce conflicting conclusions. In contrast, ML approaches that incorporate explicit linguistic markers, such as specific metrics for lexical diversity, production complexity, and grammatical fragmentation, offer a more systematic and reliable evaluation. This explicit method ensures that every decision is based on predefined, objective features, reducing the ambiguity inherent in free-form, zero-shot reasoning and ultimately enhancing classification accuracy.

There were also cases where Llama-3-8b's zero-shot prompting successfully distinguished between language that indicates PCI and language that does not. In a Facebook post with disorganized language, Llama-3-8b responded "The post is filled with fragmented sentences and inconsistent references in time, such as 'I can't tell if today is the day or yesterday was, everything's a blur'. Based on these observations, I would classify this as 1, indicating probable cognitive impairment". Here, the model correctly picks up on the disjointed narrative and temporal confusion as signals of cognitive decline. In another Facebook post with coherent language, Llama-3-8b responded "The post is well-organized and the sentence structure is clear, for example: 'I enjoyed my day at the park, and I feel grateful for the wonderful experiences I had'. Based on this clarity, I would classify this as 0, meaning no signs of cognitive impairment". In this case, the model appropriately identifies the coherent, logical structure of the language as evidence against cognitive impairment.

To sum up, our zero-shot prompting experiments suggested that without explicit linguistic markers, LLMs were promising but not robust. This indicated that with raw social media posts, current AI systems could detect some amounts of PCI. Adding overt structured linguistic indices

Table 6
Cohen's d measurement of stability of linguistic features across modality.

Feature	HC (DementiaBank vs. Facebook posts)	PCI (DementiaBank vs. Facebook posts)
VADER-compound	0.133	0.48
Valence	-0.092	-1.04
Arousal	0.379	1.401
Cosine similarity	0.782	1.566
PRON%	-0.716	-2.135
NOUN%	1.135	0.859
Noun_to_Verb	0.812	0.329
Pron_to_Noun	-1.83	-1.85
Noun_to_Adj	-0.556	-0.526
Verb_to_Adj	-1.649	-0.606
TTR	1.294	0.339
SemD	0.918	-0.309
Dependents per clause	0.621	1.404
Coordinate phrases per clause	0.648	1.54
Complex nominals per clause	1.31	1.691
Total dependents	-0.886	0.178
Total complex nominals	0.085	0.437
Mean GPT2 surprisal scores	-0.364	-0.81
Median Mistral surprisal scores	-3.573	-12.736

improved LLMs' efficacy in teasing apart PCI from HC, leading to more sensitive and robust predictions. Although taking pre-trained LLMs off-the-shelf seemed end-to-end and promising (based on qualitative analyses), linguistic biomarkers showcased more robustness in systematically revealing the cognitive status of the social media posts' producers. Furthermore, we highlight the challenge of LLM prompting in social media posts. We closely examined cases where the model failed to classify posts as PCI, even though clustering initially labeled them as such. We suspect these cases are genuinely ambiguous, where many posts involve both caregivers and their loved ones, making it unclear which parts were actually produced by the person with PCI. This ambiguity may explain why the LLM struggled to classify them correctly using only zero-shot prompting.

On the other hand, for structured prompting, quantitatively, results showed that Llama-3-8b achieved an accuracy of 0.804. It was able to precisely label most of the PCI cases (precision = 0.901), and it showed fine performance in catching all the PCI cases (recall = 0.685). When detecting PCI, Llama-3-8b with structured prompting outperformed zero-shot prompting in accuracy and precisely labeling PCI, but still it was not as robust as most of the predictive models involving explicit structured linguistic markers. This also resonates with recent findings about the linguistic blind spots of LLMs, where even some of the most capable LLM makes (consistent) errors in basic NLP tasks such as detecting linguistic structures, identifying embedded clauses, and recognizing verb phrases (Cheng and Amiri, 2025).

Qualitatively, with decision criteria specified in the prompt variant, we did observe cleaner output from Llama-3-8b, where it followed the instruction and only output a number of 1 or 0 without explaining intermediate reasoning steps. For the few cases where Llama-3-8b output the explanation, for example, *"Based on the linguistic evidence, I classify this social media post as HC (healthy control) and return 0. The post exhibits a positive and encouraging tone, with a focus on promoting action and ... The language is clear and concise, with a good balance of nouns, verbs, and adjectives. The sentence structure is varied, with a mix of short and longer sentences. The post also uses rhetorical questions to engage the reader and encourage participation. There is no evidence of increased negative sentiment, reduced positive sentiment, or high semantic similarity due to redundancy or lexical repetition. The part-of-speech patterns, lexical diversity, and syntactic complexity are all within normal ranges for a healthy control post..."*, Llama-3-8b's reasoning appeared to be consistent, and its output aligned with the post's true label. Overall, this prompt variant suggests that a more structured prompt encoding the relevant linguistic criteria could help LLM make informed decisions.

4. Discussion

Natural language analysis has been recently applied to quantify aging-related symptom severity through linguistic biomarkers in clinical settings. Yet, no attention has been paid to examine whether these markers can be robustly applied to social media text. This study is among the first to investigate the extent to which linguistic biomarkers can reliably detect early signs of cognitive impairment in social media posts, with some biomarkers proving more interpretable, stable, and effective than others. Here, we delve into our key findings on the robustness of linguistic markers. We break down robustness into three distinct aspects: markers' stability, interpretability, and efficacy. We also explain the (in-)consistencies between these markers and our predictions, as well as previous findings in the areas of modality difference, data production and collection, and the computation of linguistic markers.

4.1. Stability and interpretability of linguistic markers for aging

Our results revealed that some linguistic markers were interpretable in the unstructured social media posts, in a sense that they behaved as predicted in both sign and magnitude, while some were less

interpretable with respect to prior findings and our predictions; and some linguistic markers were stable across clinical and social media posts datasets, while some were less stable (c.f., Figs. 1 and 2 and Table 5).

Consistent with prior work, we found that lexical diversity, particularly type-token ratio (TTR), was among the most robust and interpretable markers of verbal skill impairment (Fraser et al., 2015, 2019; Yancheva et al., 2015; Beltrami et al., 2018). We extend this finding beyond the structured clinical transcripts typically analyzed in previous studies by testing it using unstructured Facebook posts. These results resonate with earlier findings that individuals with cognitive impairment tend to exhibit a restricted vocabulary and repetitiveness in their language, reflecting underlying deficits in lexical access and flexibility, even in their social media communications.

In terms of sentiment analysis, our work resonates with prior studies showing that individuals with (mild) cognitive impairment are more likely to use negative language and express lower affective valence (Scibelli, 2018; Morales et al., 2017; Lam et al., 2014; McIntyre et al., 2015; Yeung et al., 2021; Lindsay et al., 2021). The relatively stable performance of valence-based sentiment features across both datasets suggests that executive dysfunction in PCI may manifest as shifts in affective tone, detectable even in brief or informal text. Importantly, these signals were particularly salient in social media posts, which often reflect personal experiences and emotions. This suggests that social media may offer a distinctive lens into affective and cognitive states, augmenting traditional assessments with more spontaneous and emotionally rich content.

Our findings on syntactic complexity and part-of-speech patterns revealed more mixed results. While some previous studies reported that cognitive impairment is associated with a decrease in syntactic complexity and an altered distribution of POS (e.g., higher pronoun-to-noun ratios, reduced clause depth) (Fraser et al., 2015, 2019; Yancheva et al., 2015; Orimaye et al., 2014; Pakhomov et al., 2010, among others), we found these markers to be less consistent across modalities. It is possible that syntactic measures are more sensitive to task structure and genre. For instance, picture description tasks used in clinical studies elicit constrained output that makes syntactic differences more visible, whereas free-form social media posts vary widely in style and content. Our results suggest that while POS and syntactic indices hold theoretical value, their diagnostic utility may depend on more structured or homogeneous linguistic inputs.

Surprisal-based features derived from LLMs provided an exploratory angle to our analysis. Prior studies suggested that surprisal increases in individuals with aphasia or other language disorders due to decreased predictability and grammatical cohesion in their speech (Rezaii et al., 2021; 2024; Cong et al., 2024). In our study, LLM-surprisal features showed only modest predictive value in social media contexts, likely due to the noisier, more variable nature of user-generated text. However, their improved performance in clinical datasets suggests their potential when applied to grammatically constrained inputs. These findings indicate the promise of surprisal-based measures. Future work needs to fine-tune such methods for application to informal, online discourse.

In summary, our results suggested that lexical diversity markers, especially TTR, were the most interpretable and stable among all the linguistic biomarkers tested. Our finding implied that a significant decrease in TTR was not only a solid linguistic marker in structured clinical data, it was also a reliable marker for detecting PCI in social media posts. Decline in text-level vocabulary diversity turned out to be a robust index of verbal skill impairment in PCI. Second, sentiment markers, especially valence, were also mostly interpretable, although not as stable as lexical diversity indices. Lower valence scores, namely negative emotions in texts, were discriminative in predicting PCI. Negative sentiment proved to be an interpretable and relatively stable proxy to impairment in executive functioning and reasoning. This is especially striking in social media posts, suggesting the distinct cognitive insights that social media platforms such as Facebook can provide,

which would otherwise be too implicit to quantitatively pinpoint.

The resonance between our findings and prior clinical work suggests the feasibility of recasting linguistic biomarkers for PCI detection in social media. Yet, our work also highlights the importance of adapting these markers to the specific characteristics of online text. Language in social media is shaped by platform norms, user intent, and genre conventions, all of which can modulate linguistic expression. The differences we observed between clinical and social media datasets suggest that while core linguistic impairments may persist across contexts, the best-performing features may differ depending on the medium. These insights point to the need for modality-aware approaches that balance clinical validity with ecological generalizability.

We present an approach that extends interpretable and stable linguistic markers beyond clinical settings, leveraging social media data to capture real-world language use among aging populations. The expansive and naturally occurring text found in online posts provides a uniquely rich resource for tracking and revealing linguistic markers, which may serve as early indicators of cognitive decline or shifts in well-being. Notably, similar ML models applied to social media narratives have yielded decent accuracy, capitalizing on the diverse contexts available in these platforms. This shift from controlled clinical assessments to the analysis of everyday social media language offers the potential to develop scalable, automated pipelines that continuously monitor cognitive health in a less intrusive, yet highly informative, manner.

By combining insights from both clinical and social media analyses, AI-driven approaches are paving the way for a more comprehensive understanding of aging. Such integrative models not only validate traditional neuropsychological assessments but also enrich them by incorporating the dynamic and nuanced language use observed in naturalistic settings. Ultimately, this interdisciplinary convergence of AI, computational linguistics, and gerontology promises to enhance early detection and intervention strategies, making it possible to support healthy aging on a much broader, community-wide scale.

4.2. Efficacy of linguistic biomarkers for PCI detection

Predictive modeling (Tables 3 and 4) as well as zero-shot prompting provided evidence that AI language models and NLP are effective and efficient to enhance gerontologists' research, particularly in PCI detection. With F1-scores ranging from 0.86 to 0.96, our findings suggested computational systems' excellent performance. Notably, the improved efficacy of predictive models incorporating linguistic biomarkers for PCI detection, compared to LLMs zero-shot without linguistic markers, further highlights the critical value of linguistic biomarkers.

There were a few linguistic markers that are notably effective across datasets (Figs. 1 and 2). Lexical diversity markers and sentiment indices ranked higher, whereas markers involving LLMs, such as surprisals and text similarity, contributed very moderately to models' predictions. Syntactic complexity measures and POS indices' efficacies were variable. It is likely that LLMs related markers require more careful and structured text format. They ranked higher and were more effective in models sourced from the clinical datasets than from the free-formatted social media posts. We also speculate that language boundedness and content control may influence the efficacy of syntactic measures (Cong et al., 2024). For example, picture description tasks, which were commonly used in clinical settings, tend to produce constrained language with limited content. In contrast, when evaluating posts on aging-related topics, syntactic complexity indices and POS markers may be more sensitive to subtle differences in content, which in turn could affect assessments of the cognitive status of the posts' producers.

Notably, semantic similarity showed contrasting patterns across datasets (decreased in clinical, increased in social media). It is likely that this inconsistency is associated with differences in repetition and content richness. Recent work on LLM-based similarity metrics might help contextualize this pattern. Cong (2024) suggested that cosine similarity

derived from contextualized word- and sentence-level embeddings can capture meaningful aspects of writing proficiency, but also revealed that similarity scores are not always straightforwardly interpretable, depending on text cohesion, repetition, and stylistic conventions. Cong (2024) found that higher similarity does not necessarily indicate lower proficiency; instead, it may reflect greater connectedness, standardization, or adherence to shared discourse norms. Applying this line of work to our findings, increased semantic similarity in social media posts may reflect thematic convergence, repeated discussion of shared experiences, or formulaic language commonly used in online communities, and not necessarily reduced linguistic competence. In contrast, clinical interviews, though more structured, may elicit more individualized narrative responses, leading to lower similarity despite greater task control. This interpretation aligns with observations from the computational semantics literature that similarity metrics are sensitive to the balance between lexical diversity and textual cohesion, and that familiarity with genre conventions can increase similarity even in otherwise proficient language use. As demonstrated in Pranav et al. (2024) and Cong (2024), different LLM architectures also vary in semantic similarity interpretation. Future work could take our pipeline and extend it to more systematic investigations of various LLMs-based similarity metrics.

Overall, our results suggest that integrating AI and computational linguistics into gerontology research is opening new avenues for assessing cognitive health and aging beyond traditional clinical environments. Our findings demonstrated that advanced machine learning techniques, such as gradient boosting and random forests classifiers, can robustly differentiate between cognitive states by analyzing linguistic patterns in not only clinical datasets but also in other modalities, such as social media posts, resonating with previous findings in Thorstad and Wolff (2019), Bach et al. (2018) and Meier et al. (2024). These classifiers leverage a range of linguistic markers, involving proxies to executive functioning and reasoning and verbal skills, to identify multi-dimensional differences in language use that quantitatively pinpoint cognitive status. Explainability methods such as SHAP values further showcased how specific features, such as type-token ratios and pronoun frequencies, contribute to predictions, offering possible explanations into the underlying cognitive processes of aging.

4.3. Relation to multimodal and knowledge-guided models

Recent advances in multimodal and knowledge-guided learning provide a useful context for situating and extending our work. For example, Yang et al. (2025) propose MKGM, a multimodal knowledge-guided framework that integrates visual features, textual priors, and graph-structured knowledge to jointly recognize bridge defects and structural information. Although developed for a computer vision inspection task, MKGM illustrates how structured domain knowledge and hierarchical label dependencies can improve recognition in complex, real-world settings. Conceptually, these highlights promising future directions for cognitive health research, for instance, incorporating structured clinical knowledge (e.g., symptom hierarchies or neuropsychological frameworks) to guide feature interaction and interpretation, while still preserving the interpretability of linguistically grounded markers.

Similarly, Li et al. (2025) introduce Sat-RIA, a retrieval-augmented multimodal framework designed to reduce hallucinations and improve interpretability in satellite commonsense conversation by leveraging expert databases, rules, and knowledge graphs. While our study does not involve retrieval-augmented generation or external expert knowledge bases, Sat-RIA highlights the broader impact of knowledge-informed AI systems in improving reliability and transparency. For future work in cognitive impairment detection, such approaches could be adapted to retrieve verified clinical or linguistic knowledge at inference time, potentially improving the robustness of LLM-based prompting when applied to noisy, unstructured social media text. This direction may be

especially valuable for mitigating hallucinations and improving explainability in real-world healthcare related language analysis.

Pertaining to our exploration of zero-shot learning, Liu et al. (2024) propose MERL, a multimodal ECG text framework that enables zero-shot medical classification enhanced by test time clinical knowledge prompting. MERL demonstrates the effectiveness of combining multimodal representation learning with LLM-driven knowledge enhancement to generalize beyond annotated training data. In contrast, our work focuses on language only data and emphasizes the transferability, stability, and interpretability of classic linguistic biomarkers across controlled clinical interviews and spontaneous social media posts. Nevertheless, MERL points to important future directions, for instance, integrating explicit clinical knowledge or prompt engineering strategies with linguistically interpretable features to further strengthen zero-shot and few-shot cognitive health assessment in naturalistic settings.

Taken together, these studies reflect a broader advancement toward multimodal, knowledge-aware AI systems that operate in complex, real-world domains. Our findings complement this trend by showing that validated, classic linguistic biomarkers (augmented by LLM-derived measures) can generalize beyond clinical environments to everyday social media language. Against this broad context of multimodal, knowledge-aware AI systems, our research supports scalable, noninvasive monitoring of cognitive health using naturally occurring text, while maintaining interpretability and theoretical grounding. Future research that combines the strengths of knowledge-guided multimodal frameworks with robust linguistic biomarkers may further enhance early detection, interpretability, and trustworthiness in AI-driven cognitive health assessment.

4.4. Limitations and future studies

More work is needed to infer age and other demographic details from social media posts. This is important because it is a critical step for crafting comparable datasets and drawing valid conclusions from online behavior. Future research could combine information from texts where age is known with social media posts that lack such details. Our current datasets are too small to operationalize meaningful inference. Additionally, future studies might systematically extend the zero-shot approach to one-shot or few-shot learning by providing example posts for both HC and PCI, or even incorporating explicit linguistic markers and clinical data into the prompting process for a more systematic comparison. Our current study lays the foundation of applying LLM prompting that future research can build upon.

Although we leveraged previous work as well as NLP preprocessing strategies to address posts authorship concerns, we acknowledge that this represents an important area for future methodological development. For example, future research should extract posts with first-person language suggestive of self-authorship and avoid overt third-person caregiver narratives where possible. Our social media data were limited to posts from widely known organizations due to platform privacy policies. While this ensured ethical compliance and access to public content, it may limit generalizability. Public organizations may use language differently than the broader PCI population, and their online communication is likely shaped by external factors such as audience awareness or editorial oversight. Future studies should evaluate the representativeness of such posts and, where possible, include data from a wider cross-section of users. Finally, cognitive status in our social media dataset was inferred based on group membership, disease-specific versus general health-related Facebook communities, rather than confirmed clinical diagnoses. While this labeling strategy aligns with prior work in digital phenotyping, it lacks the diagnostic precision of clinical studies. Misclassification remains a possibility, particularly given the presence of undiagnosed individuals in control groups or caregiver-authored posts in PCI groups. Future research should seek to infer cognitive status through integrating clinical records, validated screening tools, or machine learning techniques specifically trained to

infer cognitive impairment from text in real-world online environments.

Despite its limitations, this study makes significant contributions to gerontology research by providing some of the first evidence of applying AI and computational linguistics to detect early signs of cognitive impairment in social media posts. Our results suggest that integrating AI and computational linguistics into gerontology research is expanding opportunities for assessing cognitive health and aging beyond traditional clinical settings. Furthermore, our findings highlight the potential value of social media posts in detecting early signs of cognitive impairment through advanced ML techniques. Given the widespread availability, the diversity and richness in content, and the minimal cost of social media data, this approach offers a scalable and accessible avenue for cognitive health assessment in aging populations.

5. Conclusion

Our study offers early evidence that linguistic biomarkers traditionally used in clinical settings can be effectively adapted to detect early signs of PCI in social media posts. Lexical diversity, especially TTR, proved to be the most interpretable, stable, and effective marker across both modalities. Sentiment markers, particularly affective valence, also showed predictive value, with negative emotional tone serving as a potential signal of cognitive decline, especially in informal online language. Markers related to syntactic complexity, POS patterns, and LLM-based surprisal measures were less consistent, likely due to their sensitivity to input structure and genre. Our findings suggest the need for modality-aware approaches that reflect the distinct linguistic features of social media. Predictive models that incorporated linguistic markers outperformed zero-shot LLM approaches, indicating the value of curated linguistic features. Limitations were acknowledged: such as small datasets, inferred cognitive labels, and uncertain authorship. Future work should prioritize more robust data collection and validation methods. Overall, our study marks a step toward real-world cognitive monitoring. By leveraging the richness of social media language, it expands opportunities for scalable, non-intrusive assessment of cognitive health, complementing traditional diagnostics and enhancing early detection strategies in aging populations.

Data availability

All the social media data in this study were drawn from Meta Platforms, Inc., (Month Accessed: September, October, and November, Year Accessed: 2024), Meta Content Library API version v5.0 <https://doi.org/10.48680/meta.metacontentlibraryapi.5.0>, accessed through the Social Media Archive (SOMAR) at Inter-university Consortium for Political and Social Research (ICPSR), an American political science and social science research consortium, based at the University of Michigan. All the clinical data in this study are drawn from the DementiaBank (<https://talkbank.org/DB/#>), Lanzi, A. M., Saylor, A. K., Fromm, D., Liu, H., MacWhinney, B., & Cohen, M. L. (2023). DementiaBank: Theoretical rationale, protocol, and illustrative analyses. *American Journal of Speech-Language Pathology*, 32(2), 426–438. <https://doi.org/10.1044/2022.ajslp-22-00281>. MacWhinney, B., Fromm, D., Forbes, M., & Holland, A. (2011). AphasiaBank: Methods for studying discourse. *Aphasiology*, 25, 1286–1307. The scripts for the analyses in this paper are available in this GitHub repository for future reproducibility: https://github.com/yancong222/NLP_text_essentials.

CRediT authorship contribution statement

Yan Cong: Writing – original draft, Writing – review & editing, Resources, Visualization, Validation, Software, Methodology, Investigation, Formal analysis, Data curation, Conceptualization. **Jingying Hu:** Writing – review & editing, Writing – original draft, Visualization, Validation, Software, Methodology, Investigation, Formal analysis, Data curation, Conceptualization. **Timothy Reese:** Writing – review &

editing, Writing – original draft, Visualization, Validation, Software, Resources, Methodology, Investigation, Formal analysis, Data curation, Conceptualization. **Hui Liu:** Writing – review & editing, Writing – original draft, Supervision, Resources, Project administration, Methodology, Investigation, Funding acquisition, Data curation, Conceptualization.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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Supplementary materials

Supplementary material associated with this article can be found, in the online version, at [doi:10.1016/j.nlp.2026.100203](https://doi.org/10.1016/j.nlp.2026.100203).

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